Série N°1 : Système monophasé :

Ex 1 :

$U_{eff} = 20V$; $T = 1ms$

1) a) $U_{max} = \hat{U} = \sqrt{2}.U_{eff} = \sqrt{2}.20 = 28,28$

b) $f = \frac{1}{T} = \frac{1}{10^{-3}} = 1000Hz = 1KHz$

2) $u(t=0) = 14,14V$

$u(t) = U_{max} \sin(\omega t + \varphi)$

$u(t=0) = U_{max} \sin(0+\varphi) \approx 14,14$

$\sin(\varphi) = \frac{14,14}{U_{max}} = \frac{14,14}{28,28} = 0,5$

$\Rightarrow \varphi = \frac{\pi}{6} = 30°$; $\omega = 2\pi f$

$= 6,28 . 1000$

$= 6280$ rad/s

Donc :

$u(t) = 28,28 \sin\left(6280t + \frac{\pi}{6}\right)$

3) $\underline{U} = U_{eff} e^{j(\omega t + \varphi)}$

$= 20 e^{j(2000\pi t + \pi/6)}$

$= 20 \left(\cos(2000\pi + \pi/6) + j \sin(2000\pi + \frac{\pi}{6})\right)$

$\underline{U} = 20 e^{j\pi/6}$

$\underline{U_1}(t) = (20 ; \frac{\pi}{2})$

4) $U_1 = \left(20 ; \frac{\pi}{2}\right)$; $U_2 = \left(40 ; \frac{\pi}{4}\right)$; $U_3 = (60 ; 0)$

$= 20 . e^{j\frac{\pi}{2}}$ $= 40 . e^{j\pi/4}$ $= 60 e^{j.0}$

$U_1 = 20j$; $U_2 = 40\left(\cos\frac{\pi}{4} + j \sin\frac{\pi}{4}\right)$

$= 20\sqrt{2} + j\, 20\sqrt{2}$;

$U_3 = 60$

$\hookrightarrow U_1 + U_2 + U_3 = (60 + 20\sqrt{2}) + j(20 + 20\sqrt{2})$

$\Rightarrow \underline{S = 88,28 + 48,28j}$

~~4) cb 2~~

$\hookrightarrow \|U\| = \|U_1 + U_2 + U_3\|$

$= \left((88,28)^2 + (48,28)^2\right)^{1/2}$

$= 100,62V$

$\varphi = \text{Arctan}\left(\frac{48,28}{88,28}\right) = 0,5005$ rad/s

$\underline{U} = 100,62 . e^{j\,0,5005}$

Ex 2 :

1) $\underline{I} = \frac{Ve}{R + jL\omega}$

$\underline{Vs} = \underline{I} . R$

$\underline{Vs} = \underline{Ve} . \frac{R}{R + jL\omega}$ (pont diviseur de tension (PDT))

$\|\underline{I}\| = \frac{Ve}{(R^2 + (L\omega)^2)^{1/2}} = \frac{230}{(100 + 100)^{1/2}}$

$= 16,26A$

$Vs = R . I = 10 . 16,26 = 162,6V$

2)

$Ve(t) = Ve \sqrt{2} \sin(\omega t + \varphi_e) \Rightarrow \underline{Ve} = Ve\, e^{j\varphi_e}$

$Vs(t) = \sqrt{2} Vs \sin(\omega t + \varphi_s) \Rightarrow \underline{Vs} = Vs\, e^{j\varphi_s}$

$i(t) = I . \sqrt{2} \sin(\omega t + \varphi_i) \Rightarrow \underline{I} = I . e^{j\varphi_i}$

$\underline{V_e} = (R + jL\omega)\,\underline{I} = \underline{Z}\,\underline{I}$

$V_e \cdot e^{j\varphi_e} = Z e^{j\varphi_z} I \cdot e^{j\varphi_i} = Z I e^{j(\varphi_z + \varphi_i)}$

$\varphi_e = \varphi_z + \varphi_i \Rightarrow \varphi_z = \varphi_e - \varphi_i$

$\varphi_z = \text{Arctg}\left(\frac{L\omega}{R}\right)$

On suppose que $\varphi_i = 0$

$\Rightarrow \varphi = \varphi_e - \varphi_i = \varphi_e = \arctan\left(\frac{L\omega}{R}\right)$

$\varphi_e = \text{Arctg}\left(\frac{10}{10}\right) \Rightarrow \varphi_e = \frac{\pi}{4}$

$\underline{V_e} = V_e \cdot e^{j\pi/4}$

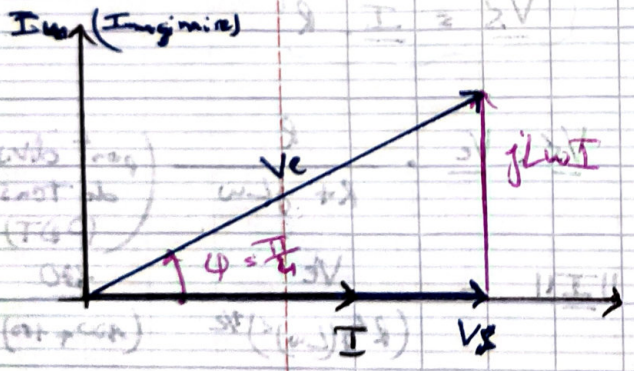
$\underline{I} = I \cdot e^{j\cdot 0}$

$\underline{V_s} = R \cdot \underline{I} = R I \cdot e^{j\cdot 0}$

$\Rightarrow v_e(t) = \sqrt{2}\, V_e \sin\left(\omega t + \frac{\pi}{4}\right)$

$i(t) = I \sin \omega t$

$V_s(t) = R \cdot I \cdot \sin \omega t$



Ex (3)

1. $I_1 = \frac{V}{R_1} = \frac{230}{20} = 11,5\ A$

2. $\underline{V} = (R_2 + jL\omega)\,\underline{I_2}$

$\underline{I_2} = \frac{\underline{V}}{R_2 + jL\omega}$

$\|I_2\| = \frac{V}{(R_2^2 + (L\omega)^2)^{1/2}} = 19,5\ A$

3. $\underline{I} = \frac{\underline{V}}{\underline{Z}}$

$\underline{Z} = R_1 // (R_2 + jL\omega)$

$= \frac{R_1(R_2 + jL\omega)}{R_1 + R_2 + jL\omega}$

$\underline{Z} = \frac{20(10 + j(20 \cdot 10^{-3} \cdot 1000\pi))}{20 + 10 + j(20 \cdot 10^{-3} \cdot 1000\pi)}$

$= \frac{200 + j\,125,6}{30 + j\,6,28}$

$I = \frac{V}{\|\underline{Z}\|} = \frac{230}{\left(\frac{200^2 + 125,6^2}{30^2 + 6,28^2}\right)^{1/2}}$

$= 29,85\ A$

4. $P = R_1 I_1^2 + R_2 I_2^2$

$= 6,44\ Kw$

$Q = L\omega I_2^2 = 2,39\ KVAR$

$S = (P^2 + Q^2)^{1/2}$

$= 6,86\ KVA$

5. $\cos(\varphi) = \frac{P}{S} = 0,93$

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Electrotech TD1 Ex4 (Suite)

1- Charge 1 :

$S_1 = 25\ KVA$; $I_1 = 100,7A$

$\cos(\varphi_1) = 0,8\ AR$ ($\varphi_1 > 0$ en arrière) $\varphi_1 = 36,9°$

$Q_1 > 0$

ou $Q_1 < 0$ $= 0,8\ AV$ (en avant)

Charge 2 :

$P_2 = 27\ KW$; $Q_2 = 36\ KVAR$

$I_2 = 195,4A$

Charge 3 :

$I_3 = 43,5A$; $P_3 = 8,66\ KW$

$\cos(\varphi_3) = 0,866\ AV$

$Q_3 < 0$

$\Rightarrow \varphi_3 < 0$

$\varphi_3 = -30,7°$

2- $P_T = 55,66\ KW$

$Q_T = 46\ KVAR$

$S_T = 72,2\ KVA$

3-

$\cos(\varphi_T) = 0,77\ AV$

$I_T = 314A$

4-

5T

$\cos\varphi' = 0,9$

$Q' = Q_T - Q_C$ ×

$\varphi' = \text{Plan}(\varphi)$ ×

$Q'_C = -cwV^2$ ×

$Q' = P_{Tot}(\varphi) - cw$

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Série 1 (Suite)

Ex4

2) $P_t = ?$, $Q_t = ?$

$P_t = \sum_{i=1}^{3} P_i = P_1 + P_2 + P_3 = 20k + 27k + 8,66k$

Donc $P_t = 55,66\ kW$

$Q_t = \sum Q_i = Q_1 + Q_2 + Q_3 = 15k + 36k - 5k$

$Q_t = 46\ kVAR$

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3) $S_t = ?$, $I = ?$, $\cos\varphi = ?$

$S_t = \sqrt{P_t^2 + Q_t^2} = 72,21\ kVA$

$I = S_t / V = \frac{72,21\ k}{230} = 313,94A$

$\cos\varphi = P_t / S_t = \frac{55,66}{72,21} = 0,77$

4- Triangle des puissance de l'ensemble des charges :

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5) $C' = ?$ $\cos\varphi' = 0,9$

charges :

$P'_t = P_t + P_c = P_t$

$Q'_t = Q_t + Q_c$

$P_t \operatorname{tg}\varphi' = P_t \operatorname{tg}\varphi - cwV^2$

$cwV^2 = P_t \operatorname{tg}\varphi - P_t \operatorname{tg}\varphi'$

AN

$$C' = \frac{P_t (tg\varphi - tg\varphi')}{\omega V^2}$$

$\varphi' > 0 \Rightarrow tg\varphi' > 0$

$\cos\varphi \rightarrow \varphi = ? \rightarrow tg\varphi = 0{,}83$

$\cos\varphi' \rightarrow \varphi' = ? \rightarrow tg\varphi' = 0{,}48$

AN $\quad C' = \dfrac{55{,}66\,10^3 (0{,}83 - 0{,}48)}{100\pi \times 230^2}$

$C' \simeq 1{,}2\ mF$

6) $C'' = ?$ $\quad \cos\varphi'' = 0{,}9$

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$P_t'' = P_t + P_{C''}^{\nearrow 0} = P_t$

$Q_t'' = Q_t + Q_{C''}$

$|Q_{C''}| > Q_t$

$\varphi'' < 0$

$Q_t'' = Q_t + Q_{C''}$

$P_t\, tg\varphi'' = P_t\, tg\varphi - C''\omega V^2$

$C'' = \dfrac{P_t (tg\varphi - tg\varphi')}{\omega V^2}$

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AN

7) Co

En pratiq

faible Va

un sur di

$\varphi'' < 0 \rightarrow tg\varphi'' < 0$

AN $\quad C'' = \dfrac{55{,}66\,10^3 (0{,}83 - (-0{,}48))}{100\pi \cdot 230^2}$

$C'' = 44\ mF$

7) $\cos\varphi' = \cos\varphi''$ mais $C'' > C'$

En pratique, on choisit le condensateur de faible valeur par économie et afin d'éviter un sur dimensionnement inutile.

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